

Referee package – SC-SCOPE: the second- and third-cumulant endpoint closure of the selection floor

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

first issued 2026-06-10 · this version issued 2026-06-12 · v1.3

1. Purpose and scope

This referee document concerns the SC-SCOPE programme: the second- and third-cumulant scope of the selection floor and its endpoint closure (sunset and quartic-difference channels). **Scope:** the all-orders endpoint at $I = 2 \times 10^{-3}$, with sign-stability certified only on the explicit window $\mathcal{W}_{\text{SC}}$ of Sec. 3. **Not claimed:** the closure is thin (sunset-limited, near-critical); it is a near-critical selection boundary, not a comfortable margin. This package does not by itself close full STEP-5B, does not close admissible-class exhaustiveness, and does not enact the full Reading-H C_{full} theorem.

2. The result

The SC-SCOPE third-cumulant joint inequality is positive at the endpoint: each channel (sunset cap $\times 1.13$, realized quartic $R_{\text{max}} = 0.385 < 0.634$ with the Parseval-pinned $1/(4\pi^2)$ convolution measure, ratio 1.0000) and their joint with the second-order cost satisfy joint > 1 (conservative $\times 1.040$, verified-floor $\times 1.082$). The endpoint sits just below the structural sunset cap; the selection floor closes (thin).

3. Structure of the argument

Joint assembly (patch v1.1-2). Under the canonical CONSERVATIVE additive bookkeeping (scscope-floor-sharpening v1.4 corrected form; NOT the retracted pairing formula), with $\Delta F_{\text{margin}} = \text{MARGIN} = 4.32 \times 10^{-3}$ (single upstream source `sectorb_common.margin_of`, $\mu^2 = 5 \times 10^{-3}$),

$$\begin{aligned} \text{joint} &= \frac{\text{MARGIN}}{C_2 + C_{\text{sunset}} + C_{\text{quartic}}}, & C_2 &= \frac{\text{MARGIN}}{\rho_{\text{lat}}}, \\ C_{\text{sunset}} &= \frac{\text{MARGIN} (1 - 1/\rho_{\text{lat}})}{S}, & C_{\text{quartic}} &= R_{\text{max}} C_2, \end{aligned}$$

equivalently $\text{joint} = [(1 + R_{\text{max}})/\rho_{\text{lat}} + (1 - 1/\rho_{\text{lat}})/S]^{-1}$, with $S = 1.13$ the rigorous sunset cap (M-ENDPOINT $M(0.33675) = 0.10495$) and ρ_{lat} the STEP-5B layer-floor ratio: conservative $\rho_{\text{lat}} = K_{\text{budget}}/K_{\text{floor}} = 266.7/40.7 = 6.55$ ($K_{\text{floor}} \leq T' \leq n_{\text{pack}}$), verified $\rho_{\text{lat}} = 12.6$ ($K_{\text{floor}} \leq 0.52 T'$). Evaluating,

$\text{joint} = \times 1.040 (\rho_{\text{lat}} = 6.55) \dots \times 1.082 (\rho_{\text{lat}} = 12.6) > 1.$

Provenance of the 0.634 threshold (patch v1.1-3). Setting $\text{joint} = 1$ at the conservative floor gives the critical quartic ratio

$$R_{\text{max}}^{\text{crit}} = \rho_{\text{lat}} \left(1 - \frac{1 - 1/\rho_{\text{lat}}}{S} \right) - 1 = 6.55 \left(1 - \frac{1 - 1/6.55}{1.13} \right) - 1 = 0.639.$$

The registered closure gate 0.634 (first registered in `scscope-floor-sharpening` v1.4 §5b, machine-checked by `scscope_joint_correction.py` 5/5) rounds this critical value DOWN, i.e. the registered gate is strictly TIGHTER than the algebraic critical value, so $R_{\max} < 0.634$ implies closure with safety in the conservative direction. The certified $R_{\max} = 0.385$ clears the gate with $\times 1.65$ headroom; the closure survives a +50% normalisation slack ($0.577 < 0.634$) but not a full factor 2 ($0.769 > 0.634$) — which is why the Parseval pin (ratio 1.0000, derived from the $(2\pi)^3$ convolution measure) is load-bearing and is itself machine-certified.

Certified sign-stability window (patch v1.1-4). The v1.0 phrase “sign-stable across nearby (I, μ^2) ” is replaced by the explicit certified window

$$\mathcal{W}_{\text{SC}} = \left\{ (I, \mu^2) : I \in [4 \times 10^{-4}, 2 \times 10^{-3}], \mu^2 \in [0.5, 2] \times \mu_0^2, \mu_0^2 = 5 \times 10^{-3} \right\},$$

on which $\text{joint} > 1$ holds throughout (evidence: `scscope_endpoint_sweep.py` 4/4, run artefact `runs/260609-scscope-endpoint-sweep/`): the I -sweep gives $\times 1.126 (4 \times 10^{-4}) \rightarrow \times 1.104 (10^{-3}) \rightarrow \times 1.040 (2 \times 10^{-3})$ (the certified endpoint is the thinnest point of the band and is positively closed; the critical boundary $\text{joint} = 1$ sits at $I \approx 2.5 \times 10^{-3}$, OUTSIDE $\mathcal{W}_{\text{SC}}$); the μ^2 -band sweep at the endpoint stays $\times 1.034\text{--}\times 1.051 > 1$, consistent with ROBUSTNESS-MU2. No sign-stability claim is made outside $\mathcal{W}_{\text{SC}}$. The thinness is structural: $\text{joint}(\rho)$ saturates at the sunset cap $\times 1.13$ as $\rho \rightarrow \infty$, so no floor sharpening can exceed it.

4. Numerical reproduction

Entry-script manifest (patch v1.1-1, amended v1.2; complete, no ellipsis). Four scripts, each self-contained with self-test asserts, all exit 0 (operator clean-run 2026-06-12: 4/4 scripts, 29/29 asserts PASS; AI-side re-run identical):

1. `codes/vacuum/scscope_ghat4_pertransfer.py` (7/7) — per-transfer quartic form factor $\widehat{G^4}(t) = (J * J)(t)$ on the realized chords.
2. `codes/vacuum/scscope_floor_sharpening.py` (5/5) — the proved floor sharpening $K_{\text{floor}} \leq T'(M)$ and $\rho_{\text{lat}} : 2.58 \rightarrow 6.55$.
3. `codes/vacuum/scscope_mendpoint_eval.py` (12/12) — M-ENDPOINT $M(0.33675) = 0.10495$ (two quadratures agree 0.61%), the rigorous sunset cap $\times 1.13$ (dressed endpoint ratio $\times 1.129 > 1$).
4. `codes/vacuum/scscope_quartic_certificate.py` (5/5) — the Parseval pin (ratio 1.0000), the certified $R_{\max} = 0.385 < 0.634$, and the certified $\text{joint} \times 1.040\text{--}\times 1.082 > 1$ (the load-bearing closure).

Supporting / historical (NOT entry scripts; not load-bearing for the closure): `codes/vacuum/scscope_joint_pairing.py` (4/4 in a full checkout) — the most-favourable joint incompatible-pairing bound, an honest-negative reference for the RETRACTED pairing formula; it reads the runtime artefacts `runs/260607-scscope-ghat4-pertransfer/result.json` and `runs/260607-scscope-joint-endpoint/result.json` and is therefore not self-contained in a bundle that omits them (operator clean-run audit 2026-06-12); `codes/vacuum/scscope_endpoint_sweep.py` (4/4) for the $\mathcal{W}_{\text{SC}}$ window evidence (Sec. 3); `codes/vacuum/scscope_joint_correction.py` (5/5) for the 0.634 gate registration. Run the entry manifest from the bundle and diff against `expected/`.

5. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) *Attack the scope.* The result is confined to the scope stated in Sec. 1; a referee may push that boundary, which is fair — the package makes no claim outside it, and the precise scope is in the footer. In particular: endpoint + the explicit window W_{SC} only; no all-intensity claim; no full STEP-5B, no exhaustiveness, no full C_{full} .
- (β) *Attack the tier / over-claim.* No prose here is stronger than the registered grade (footer no-overclaim line); the status is PUBLISHED-BUNDLE CONFIRMED as a T5 thin-certified support package (bundle SC-SCOPE-T5-260612; NOT a T6/T7 promotion), and the claim-ledger tier is the arbiter. The closure is explicitly flagged THIN and sunset-limited.
- (γ) *Attack reproduction / self-containment.* Each reproduction script carries self-test asserts and ships in the folder’s bundle with its transitive dependencies (imports and runtime file reads, resolver v1.3.0); a referee runs them and diffs against `expected/`.

Quantitative sanity check (mandatory): the 0.634 gate is re-derived from its inputs ($R_{max}^{crit} = 6.55(1 - (1 - 1/6.55)/1.13) - 1 = 0.639$, registered DOWN to 0.634, safe direction); the joint is re-assembled from its terms ($[(1 + 0.385)/6.55 + (1 - 1/6.55)/1.13]^{-1} = 1.040$); the falsifier (footer) is concrete and pre-registered; and every numerical value cited is reproduced by the 4-script entry manifest (29/29 asserts PASS, exit 0; operator clean-run 2026-06-12 identical).

6. Result footer

Result ID:	B5-BEYOND-LAYER-BOUND / SC-SCOPE referee package
Precise statement:	At the all-orders endpoint $I=2e-3$ the SC-SCOPE third-cumulant joint > 1: realized quartic $R_{max}=0.385<0.634$ (Parseval-pinned $1/(4\pi^2)$ convolution measure, ratio 1.0000), sunset cap $x1.13$, joint = $MARGIN/(C_2+C_{sunset}+C_{quartic}) = [(1+R_{max})/\rho + (1-1/\rho)/1.13]^{-1} = x1.040$ ($\rho_{lat}=6.55$) .. $x1.082$ ($\rho_{lat}=12.6$) > 1 (thin, sunset-limited, near-critical). Sign-stable on the certified window $W_{SC} = [4e-4, 2e-3] \times [0.5, 2]\mu O^2$ only.
Scope:	SC-SCOPE all-orders endpoint selection floor + the explicit window W_{SC} . Thin (sunset-limited near-critical). NOT all-intensity; NOT full STEP-5B; NOT exhaustiveness; NOT full C_{full} .
Evidence grade:	third-cumulant endpoint closure, thin-certified (T5); EXECUTED (4 entry scripts, 29/29 asserts PASS, exit 0; operator clean-run 4/4 PASS 2026-06-12).
Reproduction command:	python codes/vacuum/scscope_ghat4_pertransfer.py ; codes/vacuum/scscope_floor_sharpening.py ; codes/vacuum/scscope_mendpoint_eval.py ; codes/vacuum/scscope_quartic_certificate.py (joint_pairing: supporting/historical only, requires full-checkout run artefacts)
Falsifier:	a recomputed joint ≤ 1 at the endpoint; or a convolution-normalisation that doubles Ghat4 ($R_{max}>0.634$); or joint ≤ 1 anywhere in W_{SC} .
Tier before / after:	PUBLISHED-BUNDLE CONFIRMED (operator CONFIRM 2026-06-12); bundle SC-SCOPE-T5-260612; T5 thin-certified endpoint support package

(NOT T6/T7; NOT full C_full). No B5 claim-tier promotion.

No-overclaim statement: the endpoint closes THIN (sunset-limited near-critical); not a comfortable all-intensity statement; sign-stability is claimed only on the explicit window W_SC.

Next required action: none for this package (PUBLISHED). Mainline: DR-2 referee package; full STEP-5B budget comparison + admissibility gates remain separately tracked.